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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: The Ethics Committee of Weifang Yidu Central Hospital approved the present research.
Consent: Written informed consent was obtained from all participants prior to surgery.
IACUC: The animal experiment was approved by the Ethics Committee of Animal Experiments at Weifang Yidu Central Hospital (Shandong , China) .
Euthanasia Agents: On day 30 after cell inoculation , the mice were euthanized by cervical dislocation , and the tumor xenografts were resected , weighed and imaged.
Inclusion and Exclusion Criteria
The inclusion criteria were as follows: i) Diagnosed with NSCLC ii) was not treated with anticancer therapies prior to surgery and iii) agreed to take part in the research.
Attrition
not detected.
Sex as a biological variable
Materials and methods Patient samples Paired NSCLC and adjacent normal tissues were collected from 57 patients (33 males, 24 females age range, 32-75 years mean age, 56.7+-7.4 years) from the Weifang Yidu Central Hospital (Weifang, China) between February 2014 and December 2015.
Subject Demographics
Age: Materials and methods Patient samples Paired NSCLC and adjacent normal tissues were collected from 57 patients (33 males, 24 females age range, 32-75 years mean age, 56.7+-7.4 years) from the Weifang Yidu Central Hospital (Weifang, China) between February 2014 and December 2015.
Randomization
The number of migrated cells in five random fields of view was counted using an inverted light microscope.
Blinding
not detected.
Power Analysis

not detected.
Replication
Statistical analysis All experiments were repeated three times and all the results are presented as the mean +- SD.
Number: Statistical analysis All experiments were repeated three times and all the results are presented as the mean +- SD.
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
The supernatant (10) was used as the input, and the rest was incubated in RIP buffer containing magnetic beads conjugated with Argonaut 2 (Ago2) or normal IgG control antibodies (EMD Millipore).	normal IgG		
Subsequently, the membranes were incubated overnight at 4degC with primary antibodies targeting HMGA1 (cat. no. ab129153 1:1,000 dilution) or GAPDH (cat. no. ab128915 1:1,000 dilution) (both from Abcam).	HMGA1		
	GAPDH		
Experimental Models: Cell Lines			
Cell culture The BEAS-2B (CRL-9609(tm)) human normal bronchial epithelial cell line was cultured in bronchial epithelial cell growth medium (Lonza Group, Ltd.) containing 10 FBS (Gibco Thermo Fisher Scientific, Inc.).	BEAS-2B		Suggestion: RRID:CVCL_0168(link)
The H522 (CRL-5810(tm)) and NCI-H460 (H460 CVCL_01459) (both from American Type Culture Collection) NSCLC cell lines were cultured in RPMI-1640 medium supplemented with 10 FBS (both from Gibco Thermo Fisher Scientific, Inc.).	NCI-H460		Suggestion: RRID:CVCL_0459(link)
Another two NSCLC cell lines, SK-MES-1 and A549, were purchased from the National Collection of Authenticated Cell Cultures and maintained in MEM and F-12K medium, respectively and both supplemented with 10 FBS (all from Gibco Thermo Fisher Scientific, Inc.	SK-MES-1		Suggestion: RRID:CVCL_0630(link)
The H460 cell line is a large cell lung cancer cell line , while the H522 and A549 cell lines are lung adenocarcinoma (LUAD	H460		Suggestion: RRID:CVCL_0459(link)
	A549		Suggestion: RRID:CVCL_0023(link)
G (both from Addgene Inc .) , were transfected into 293T cells (The National Collection of Authenticated Cell Cultures) using Lipofectamine(r) 2000 (Invitrogen Thermo Fisher Scientific , Inc .) .	293T		Suggestion: RRID:CVCL_0063(link)
Experimental Models: Organisms/Strains			

Male BALBc nude mice (mean weight +- SD , 20.4+-1.1 g) , aged 4-6 weeks , were purchased from Beijing HFK Bioscience (Beijing , China) .	BALBc nude		
Recombinant DNA			
High mobility group AT-hook 1 (HMGA1)-overexpression vector pcDNA3.1-HMGA1 and empty pcDNA3.1 control was obtained from GenScript .	HMGA1)-overexpression		
	pcDNA3.1-HMGA1		
	pcDNA3.1		Suggestion: RRID:Addgene_100544(link)
LINC01748 short hairpin RNA (shRNA sh-LINC01748) and non-targeting scramble shRNA (sh-NC) were purchased from Shanghai GenePharma Co . , Ltd. , and inserted into the pLKO.	pLKO		Suggestion: RRID:Addgene_21964(link)
The resulting vectors , alongside psPAX2 and pMD2 .	psPAX2		
	pMD2		Suggestion: RRID:Addgene_12259(link)
G: pLKO.1) was 1:1:2 .	pLKO.1		Suggestion: RRID:Addgene_8453(link)
Luciferase reporter assay The HMGA1 3-UTR and LINC01748 fragments with the wild-type (WT) miR-520a-5p-binding site were prepared by Shanghai GenePharma Co., Ltd., and inserted into the psiCHECK2 luciferase reporter vector (Promega Corporation), yielding the HMGA1-WT and LINC01748-WT reporter vectors.	psiCHECK2		
Software and Algorithms			
CellQuest software v.2.9 (BD Biosciences) was used for data analysis .	CellQuest		
ENCORI and TargetScan (http:www.targetscan.org), bioinformatic tools were used to analyze miRNAs and identify the downstream targets of miR-520a-5p.	TargetScan		Suggestion: TargetScan (RRID:SCR_010845)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Significant differences between two groups were analyzed using a paired or unpaired Students t-test.	t-test
Comparisons between multiple groups were analyzed using one-way analysis of variance followed by a Tukeys post hoc test.	analysis of variance
	Tukeys post hoc test
Pearsons correlation analysis was used to detect the correlation between gene expression levels.	Pearsons correlation
Kaplan-Meier analysis and the log-rank test were utilized for analysis of overall survival time.	log-rank test
Oligonucleotides	
The si-LINC0174811 sequence was 5-GTC ATT TTA AGC TAT TTA ATT TT-3 the si-LINC0174812 sequence was 5-ATG TAT TTA CAT CGT TTT ATA CA-3 and the si-NC sequence was 5-CAC GAT AAG ACA ATG TAT TT-3 .	5-GTC ATT TTA AGC TAT TTA ATT
	5-ATG TAT TTA CAT CGT TTT ATA CA-3
	5-CAC GAT AAG ACA ATG TAT TT-3
The miR-520a-5p mimic sequence was 5-UCU UUC AUG AAG GGA GAC CUC-3 and the NC mimic sequence was 5-UUG UAC UAC ACA AAA GUA CUG-3 .	5-UCU UUC AUG AAG GGA GAC CUC-3
	5-UUG UAC UAC ACA AAA GUA CUG-3
The miR-520a-5p inhibitor sequence was 5-AGA AAG UAC UUC CCU CUG GAG-3 and the NC inhibitor sequence was 5-ACU ACU GAG UGA CAG UAG A-3 .	5-AGA AAG UAC UUC CCU CUG GAG-3
	5-ACU ACU GAG UGA CAG UAG A-3
The following primers were used: LINC01748 forward , 5-TGC TAC ATG GGT AAG GGA GGG-3 and reverse , 5-TTC TGG TAG ACG CAC TCG TCC T-3 HMGA2 forward , 5-CGC ACT TCA GCC CAG GGA CAA-3 and reverse , 5-CTT CAG CCC AGG GAC AAC GGT CTC TTA GGA GAG GGC TCA C-3 GAPDH forward , 5-CGG AGT CAA CGG ATT TGG TCG TAT-3 and reverse , 5-AGC CTT CTC CAT GGT GGT GAA GAC-3 miR-520a-5p forward , 5-TCG GCA GGC UCC AGA GGG AAG U-3 and reverse , 5-CAC TCA ACT GGT GTC GTG GA-3 and U6 forward , 5-CTC GCT TCG GCA GCA CA-3 and reverse , 5-AAC GCT TCA CGA ATT TGC GT-3 .	5-TGC TAC ATG GGT AAG GGA GGG-3
	5-TTC TGG TAG ACG CAC TCG TCC T-3
	5-CGC ACT TCA GCC CAG GGA CAA-3
	5-CTT CAG CCC AGG GAC AAC GGT CTC TTA GGA GAG GGC TCA
	5-CGG AGT CAA CGG ATT TGG TCG TAT-3
	5-AGC CTT CTC CAT GGT GGT GAA GAC-3
	5-TCG GCA GGC UCC AGA GGG AAG U-3
	5-CAC TCA ACT GGT GTC GTG GA-3
	5-CTC GCT TCG GCA GCA CA-3
	5-AAC GCT TCA CGA ATT TGC GT-3
The sh-LINC01748 sequence was 5-CCG GGT CAT TTT AAG CTA TTT AAT TTT CTC GAG AAA ATT AAA TAG CTT AAA ATG ACT TTT TG-3 and the sh-NC sequence was 5-CCG GCACGA TAA GAC AAT GTA TTT CTC GAG AAA TAC ATT GTC TTA TCG TGT TTT TG-3 .	5-CCG GGT CAT TTT AAG CTA TTT AAT TTT CTC GAG AAA ATT AAA TAG CTT AAA ATG ACT TTT TG-3
	5-CCG GCACGA TAA GAC AAT GTA TTT CTC GAG AAA TAC ATT GTC TTA TCG TGT TTT TG-3

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